Decent Coaching Classes

10th Semi

Std X | Maths Part II | Chapter 1: Similarity | Worksheet

Section A: Ratio of Areas and Basic Proportionality Theorem

1. The base of a triangle is 9 and its height is 5. The base of another triangle is 10 and its height is 6. Find the ratio of their areas.

2. Two triangles have equal heights. Their bases are 8 cm and 12 cm. Find the ratio of their areas.

3. In ΔABC, point D lies on side BC such that BD = 6 and DC = 9. Find:

(i) $A(\Delta ABD) : A(\Delta ADC)$

(ii) $A(\Delta ABD) : A(\Delta ABC)$

4. In ΔABC, point D lies on AB and point E lies on AC such that $DE \parallel BC$. If AD = 3, DB = 6 and AE = 2.5, find EC.

5. In ΔPQR, point S lies on PQ and point T lies on PR. PS = 4, SQ = 6, PT = 6 and TR = 9. Is $ST \parallel QR$? Give reason.

Section B: Angle Bisector Theorem, Parallel Lines and Tests of Similarity

6. In ΔABC, ray AD bisects $\angle A$ and D lies on BC. If AB = 6, AC = 9 and BD = 4, find DC.

7. In ΔPQR, ray RS bisects $\angle R$ and S lies on PQ. If PR = 10, RQ = 15 and PS = 6, find SQ.

8. Three parallel lines are cut by two transversals. On the first transversal the intercepts are 4 cm and 6 cm. On the second transversal the intercepts are 6 cm and x cm (in the same order). Find x.

9. In ΔABC and ΔPQR, $\angle A = \angle P = 50°$ and $\angle B = \angle Q = 70°$.

(i) Find $\angle C$ and $\angle R$.

(ii) Are the triangles similar? Name the test and write the similarity in correct correspondence.

10. In ΔABC, AB = 3, BC = 4, AC = 5. In ΔPQR, PQ = 6, QR = 8, PR = 10. Are the triangles similar? Name the test and write the similarity.

Section C: Areas of Similar Triangles and Applications

11. The ratio of the corresponding sides of two similar triangles is 3 : 5. Find the ratio of their areas.

12. $\Delta ABC \sim \Delta PQR$. If $A(\Delta ABC) = 16$ and $A(\Delta PQR) = 25$, find AB/PQ.

13. The areas of two similar triangles are 36 sq. cm and 100 sq. cm. If a side of the smaller triangle is 9 cm, find the corresponding side of the bigger triangle.

14. A 6 m tall pole casts a shadow of 4 m. At the same time, a tree casts a shadow of 10 m. Find the height of the tree.

15. ΔABC and ΔDEF are equilateral triangles with $A(\Delta ABC) : A(\Delta DEF) = 1 : 4$. If AB = 5 cm, find DE.

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Answer Key and Solutions

Section A

1. Ratio = $(9 \times 5) : (10 \times 6) = 45 : 60 = 3 : 4$

2. Heights are equal, so areas are proportional to bases.
Ratio = $8 : 12 = 2 : 3$

3. BC = 6 + 9 = 15. Triangles ABD, ADC, ABC have the same height from A.
(i) $A(\Delta ABD) : A(\Delta ADC) = BD : DC = 6 : 9 = 2 : 3$
(ii) $A(\Delta ABD) : A(\Delta ABC) = BD : BC = 6 : 15 = 2 : 5$

4. DE ∥ BC, so $AD/DB = AE/EC$ (Basic Proportionality Theorem)
$3/6 = 2.5/EC \rightarrow EC = (6 \times 2.5)/3 = 5$

5. $PS/SQ = 4/6 = 2/3$ and $PT/TR = 6/9 = 2/3$
$PS/SQ = PT/TR$, so ST ∥ QR (converse of Basic Proportionality Theorem). Yes.

Section B

6. $AB/AC = BD/DC$ (angle bisector theorem)
$6/9 = 4/DC \rightarrow DC = (9 \times 4)/6 = 6$

7. $PR/RQ = PS/SQ$ (angle bisector theorem)
$10/15 = 6/SQ \rightarrow SQ = (15 \times 6)/10 = 9$

8. Intercepts on transversals by parallel lines are in proportion:
$4/6 = 6/x \rightarrow x = (6 \times 6)/4 = 9$ cm

9. (i) $\angle C = 180° - 50° - 70° = 60°$; $\angle R = 180° - 50° - 70° = 60°$
(ii) $\angle A \cong \angle P$ and $\angle B \cong \angle Q$, so $\Delta ABC \sim \Delta PQR$ by AA test.

10. $AB/PQ = 3/6 = 1/2$, $BC/QR = 4/8 = 1/2$, $AC/PR = 5/10 = 1/2$
All three ratios are equal, so $\Delta ABC \sim \Delta PQR$ by SSS test of similarity.

Section C

11. Ratio of areas = $(3)^2 : (5)^2 = 9 : 25$

12. $A(\Delta ABC)/A(\Delta PQR) = AB^2/PQ^2$
$16/25 = AB^2/PQ^2 \rightarrow AB/PQ = 4/5$ (taking square roots)

13. Ratio of areas = 36/100, so ratio of sides = $6/10 = 3/5$
$9/x = 3/5 \rightarrow x = (9 \times 5)/3 = 15$ cm

14. The pole with its shadow and the tree with its shadow form similar triangles (AA test).
$6/4 = h/10 \rightarrow h = (6 \times 10)/4 = 15$ m

15. Equilateral triangles are similar. $A(\Delta ABC)/A(\Delta DEF) = AB^2/DE^2 = 1/4$
$AB/DE = 1/2 \rightarrow 5/DE = 1/2 \rightarrow DE = 10$ cm